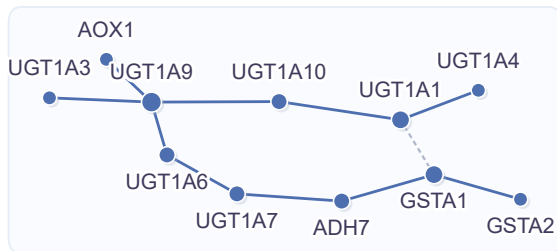
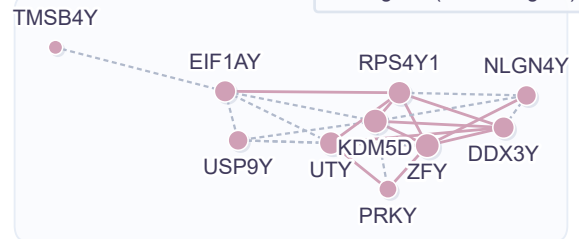


a)

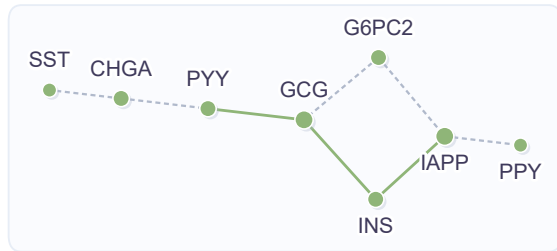
## Drug-metabolising enzymes



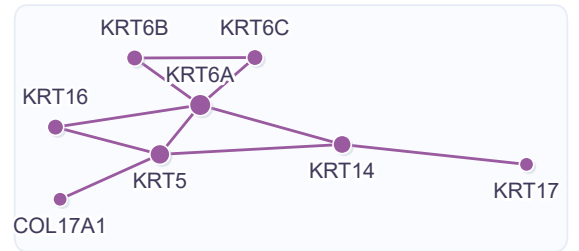
## Y chromosome



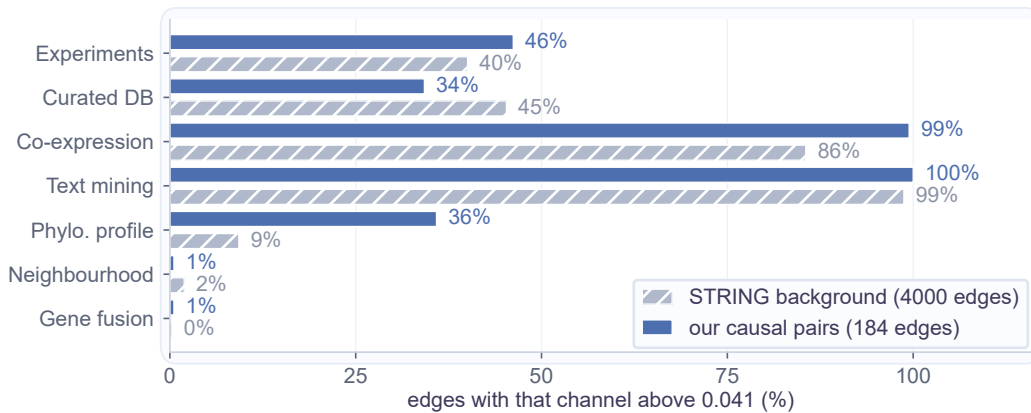
## Endocrine pancreas



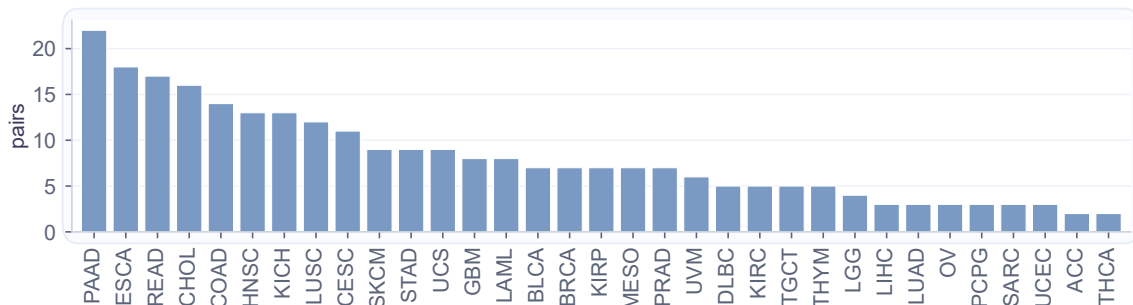
## Basal keratinocytes



b)



c)



STRING v12 (score  $\geq 0.7$ , human): 184 of 2286 unique causal pairs, 266 of 2445 directed edges (8.0%).

Random pairs from the same 1775 genes reach that score in 0.25% of cases, a 31.7-fold enrichment (Fisher  $p = 8e-206$ ).

Solid edges carry experimental or curated evidence, dashed edges co-expression only (61 of 184).

Four largest of 87 connected components; the rest are 131 edges including 60 two-gene pairs.